Statistics: Type I and Type II Errors

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DIRECTIONS *For each scenario, identify the type of error or compute the requested value. Use: Type I Error = reject a true H■ (α); Type II Error = fail to reject a false H■ (β); Power = 1 – β.*

1 Identify the type of error:

$$H_0 \text{ is true, but we reject } H_0$$

Answer: ________________________

2 Identify the type of error:

$$H_0 \text{ is false, but we fail to reject } H_0$$

Answer: ________________________

3 Find the power of the test given \beta:

$$\beta = 0.20$$

Answer: ________________________

4 Identify the error: A drug has no effect, but the test says it does.

$$H_0 : \text{drug has no effect (true)}$$

Answer: ________________________

5 Identify the error: A disease is present, but the test says it is not.

$$H_0 : \text{no disease (false)}$$

Answer: ________________________

6 If \alpha = 0.05, what is P(Type I Error)?

$$\alpha = P(\text{Type I Error})$$

Answer: ________________________

7 Which error is also called a "false positive"?

$$\text{False Positive} = \text{Reject true } H_0$$

Answer: ________________________

8 Which error is also called a "false negative"?

$$\text{False Negative} = \text{Miss a real effect}$$

Answer: ________________________

Answer Key & Solutions

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TEACHER NOTES *Type I (α) = false positive: reject true H■. Type II (β) = false negative: miss a real effect. Power = 1–β. Decreasing α increases β and lowers power (trade-off). Larger samples increase power.*

1 Identify the type of error:

$$H_0 \text{ is true, but we reject } H_0$$

= $$\text{Type I Error } (\alpha)$$

Rejecting a true null hypothesis is a false positive — Type I Error.

2 Identify the type of error:

$$H_0 \text{ is false, but we fail to reject } H_0$$

= $$\text{Type II Error } (\beta)$$

Failing to reject a false null hypothesis is a false negative — Type II Error.

3 Find the power of the test given \beta:

$$\beta = 0.20$$

= $$\text{Power} = 1 - \beta = 0.80$$

Power = 1 - P(Type II Error). Higher power means better ability to detect a true effect.

4 Identify the error: A drug has no effect, but the test says it does.

$$H_0 : \text{drug has no effect (true)}$$

= $$\text{Type I Error}$$

H0 is true (no effect), but we rejected it — this is a false positive.

5 Identify the error: A disease is present, but the test says it is not.

$$H_0 : \text{no disease (false)}$$

= $$\text{Type II Error}$$

H0 is false (disease exists), but we failed to reject it — false negative.

6 If \alpha = 0.05, what is P(Type I Error)?

$$\alpha = P(\text{Type I Error})$$

= $$P(\text{Type I Error}) = 0.05$$

By definition, alpha is the probability of committing a Type I Error.

7 Which error is also called a "false positive"?

$\text{False Positive} = \text{Reject true } H_0$

= Type I Error

Type I Error: we claim an effect exists when it does not.

8 Which error is also called a "false negative"?

$\text{False Negative} = \text{Miss a real effect}$

= Type II Error

Type II Error: we miss a real effect — fail to reject a false H0.

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